

Style Transfer: A Decade Survey

A Summary of Zhang & Tang (2025), arXiv:2506.19278

The problem

Every good system starts with a clear problem, and this one is easy to state. You have a content image I_c , the thing you want to depict, and a style image I_s , the way you want to depict it. Your job is to produce a third image that keeps the structure of the first and the appearance of the second. Written out, you need a function

$$\mathcal{F} : (I_c, I_s) \longrightarrow I_t.$$

That is the entire discipline in one line. The interesting part, as always, is not the goal but the constraints. Over the past decade researchers built dozens of ways to construct $\mathcal{F}$, and Zhang and Tang’s 2025 survey—a review of more than five hundred papers—is a map of every route they tried, what each one cost, and where each one broke.

Where style hides

The first breakthrough came from Gatys, who found where style lives. Run an image through a fixed recognition network and read the feature maps $\Phi_\ell(I)$ at layer ℓ . Style is not in the features themselves. It is in how they correlate. You capture that with the Gram matrix,

$$G_\ell(I) = \Phi_\ell(I) \Phi_\ell(I)^\top.$$

Match the Gram matrices of your output to those of the style image and you match the style. It works. It is also expensive: the matrix holds $\mathcal{O}(C^2)$ numbers per layer, and the original method reached the target by slow iterative optimization, grinding on the pixels for minutes per image.

Huang and Belongie found the cheaper path. You do not need every correlation. You need two numbers per channel, the mean and the standard deviation. Adaptive Instance Normalization strips the content features down and rebuilds them with the style’s statistics,

$$\text{AdaIN}(x_c, x_s) = \sigma(x_s) \frac{x_c - \mu(x_c)}{\sigma(x_c)} + \mu(x_s).$$

This costs $\mathcal{O}(C)$ instead of $\mathcal{O}(C^2)$, runs in a single forward pass, and—here is the part that matters—usually looks just as good. Usually. The survey is clear-eyed about the gap: dropping the cross-channel correlations costs you fidelity on complicated styles. Every shortcut has a price. This system has a lot of shortcuts, and the survey’s real value is in naming each price.

The whole apparatus trains against a small set of losses, all read through the same frozen network. One term preserves content, one matches style (Gram or moments, your choice), one keeps the result smooth,

$$\mathcal{L}(I_t) = \alpha \mathcal{L}_{\text{content}} + \beta \mathcal{L}_{\text{style}} + \gamma \mathcal{L}_{\text{TV}}.$$

The weights α, β, γ are the control panel. Turn them and you slide between a faithful photo and a heavy painting. There is no free setting. Push one and you lose another. That tension is the system’s core rule, and it never goes away.

The generative orders

Where the survey earns its length is in cataloguing what came next: the generative models that learned to build images from scratch, then were retooled for style. Each is its own discipline, with its own mechanics and its own failure mode.

Variational autoencoders compress an image into a latent code and reconstruct it, trained to keep that latent space smooth,

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_q[\log p(x | z)] - D_{\text{KL}}(q(z | x) \| p(z)).$$

The reward is a space you can navigate: blend two styles, interpolate cleanly. The cost is softness, since VAEs tend toward the blurry.

Generative adversarial networks pit two models against each other, a generator forging images and a discriminator trying to catch it,

$$\min_G \max_D \mathbb{E}_x[\log D(x)] + \mathbb{E}_z[\log(1 - D(G(z)))].$$

Balanced, the results are sharp and convincing. Unbalanced, the generator collapses to a single trick. Realism, bought with instability.

Flow-matching learns a velocity field that carries noise smoothly into data. *Autoregressive models* generate an image one token at a time, each conditioned on the last, which buys precise structural control at the cost of slow sequential decoding,

$$p(x) = \prod_i p(x_i | x_{<i}).$$

And *diffusion*, the paradigm that now dominates, works by reversal. Corrupt an image with noise across many steps,

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I),$$

then train a network to undo the corruption, step by step. Learn the reverse process and you can generate almost anything, guided by a style reference or a text prompt, at high resolution. The cost is compute: diffusion is heavy and slow, each denoising step touches the whole image, and fine local edits fight the global process. The survey pairs it against autoregressive models and finds them complementary—autoregression is better at layout and structure, diffusion at texture and flexible guidance—which points, predictably, toward hybrids.

How you grade it

How do you score a stylized image? There is no single number for it, and the survey does not pretend otherwise. Instead it lays out a structured rubric: perceived fidelity, content preservation, diversity, perceptual quality, and a deep-network alignment measure. Then it adds a second, more practical set: artistic merit, visual quality, computational efficiency, robustness, multimodal capability, and ethics. That last one matters more every year, as these tools grow good enough to imitate real artists and real faces.

Where it gets used

The techniques spread fast. Portraits became cartoons and caricatures. Video was restyled frame by frame while keeping motion coherent. Three-dimensional models and human animation were restyled. Text was rewritten into new tones, polite or formal or someone else's. Whole domains were adapted, so a model trained on clear conditions could still work in fog or rain. All of it needed data, and the survey inventories the archives the field depends on—Danbooru for anime, WikiArt for paintings, FFHQ and MetFaces for faces—the raw material every one of these systems learns from.

The avalanche, and what is left

Read end to end, the decade has the shape of an avalanche. A single insight, that a recognition network already separates content from style, triggered feed-forward methods, then adversarial training, then diffusion and autoregressive systems that render at scale with fine control. Each advance built directly on the last, and the pace only increased.

But the mountain is not cleared, and the survey ends on the constraints, which is the honest way to end. Diffusion and GANs remain too costly for real-time use on ordinary hardware. Vision and language still align imperfectly. No single paradigm does everything, so the future likely belongs to hybrids that borrow structure from one and texture from another.

Beneath all of it sits the law the whole field obeys: you cannot maximize fidelity, speed, and diversity at once. Improve one and you pay in the others. Ten years of work did not break that rule. It mapped it—precisely, method by method—so the next decade knows exactly where the walls are, and, if the pattern holds, exactly which ones to push on first.